PRESEMO Feedback

I like …

12:25 » The framing of the lecture and how easy concepts are made to understand. Also, the interactive nature of the lecture makes it interesting. ☆

13:54 (2.9) » Very good explanation of the structure of the course and the assignments ☆

13:54 (2.9) » The lecture goes well beyond. ☆

13:53 (2.9) » The fact of using jupyter, i think it s relevant ☆

13:53 (2.9) » How everything was explained very well and detailed before everything, and the course started with a strong explanation of why. +1 ☆

I wish …

12:25 » Nothing, the lecture was organized well. ☆

13:56 (2.9) » For this first this first lecture everything was very organised. ☆

13:55 (2.9) » The materials for the next lecture would be already available +1 ☆

Quiz Feedback

- All learning objectives were met, either independently or with some support
- The only objective with a few "not there yet" responses was distinguishing between a target variable and possible predictors based on the analytical goal.

RECAP: Modeling – variables of interest

- **The purpose of the model – how to use it and interpret it – depend on the modeler's (your) mindset**

- For (predictive) modelling
  - Target variable (Label) – what we want to predict from the data set
  - All the other variables can be used as predictors
  - Choice of target and predictors depends on your goal
    - Example: Predicting propulsion shaft power using relevant variables available in the *Cruiseships* database

Where we are on the course

We go at a pace that makes sense and adjust the depth/breadth of topics

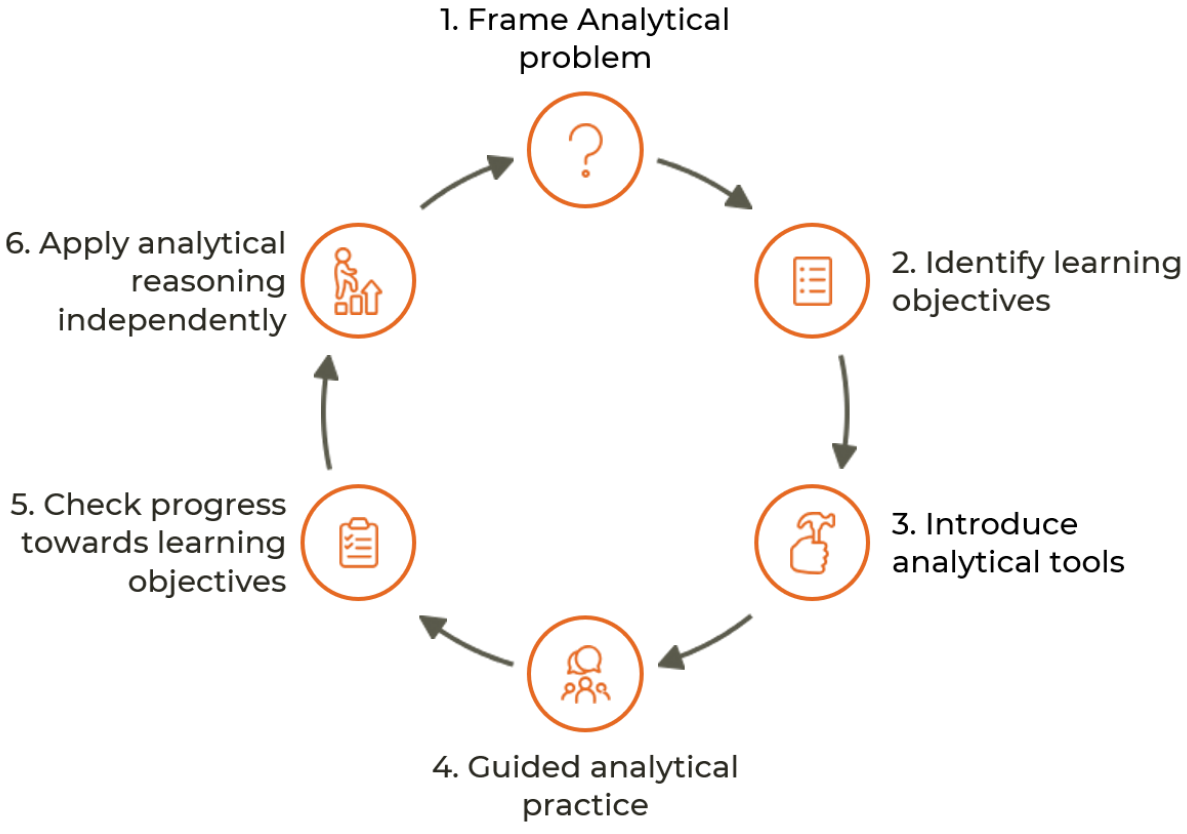
Period I

- L1: Course orientation and foundations of marine data analysis
- **L2: Exploratory Data Analysis (EDA)**
- L3 & L4: Data pre-processing
- L5: From engineering questions to analytical models
- L6: Marine data analysis in practice

Period II

- L7: Regression modelling, evaluation, and refinement
- L8 & L9: AIS data analytics
- L10 & L11: Cluster analysis
- L12: AI-assisted qualitative data analysis
- Project demonstration

Problem based learning cycle



Aalto-yliopisto
Insinööritieteiden
korkeakoulu

Exploratory data analysis

MEC-E2016 Marine Data Analysis

Reference: Contents of this lecture are partially covered in

1. Data mining: Concepts and techniques. Chapter 2.

Problem of the week – Assignment 1

Assume that you have secured an internship at Helsinki Shipyard and joined its research team. Your first task is to model propulsion shaft power using relevant variables available in the Cruiseships database. Before beginning the modelling, you need to determine whether the data are ready and which variables may be useful. Use exploratory data analysis (EDA) to investigate the following:

Grade 1 to 2: Data quality: Identify up to three important data-quality issues, such as missing values, unsuitable data types, inconsistent categories, or potentially unusual values. Explain why each may require attention.

Grade 3 to 4: Variation in the variables: Examine how the variables are distributed across the dataset. Explain the main patterns you observe, including their typical values, variability, and any notable skewness, imbalance, or unusual observations.

Grade 5: Variables relevant to propulsion shaft power: Investigate how propulsion shaft power relates to the other variables in the dataset. Also examine whether some variables are closely related to each other and may therefore provide similar information. Based on these relationships, identify which variables should be considered when modelling shaft power and explain your reasoning.

Present and interpret your findings in the report, supported by appropriate numerical summaries and/or visualizations. Use a Jupyter Notebook or another appropriate tool of your choice to perform the analysis.

Analysis is tempting, BUT is your data ready?

- The data collection can often be loosely controlled resulting in
  - Inaccuracies (out-of-range values, impossible data combinations)
  - Missing values
- Data from different sources may be in various formats, in different units, or in different conventions
- Different columns of numerical data may have very different ranges, and direct comparison is often not meaningful
- Columns containing continuous values (real numbers) can have too many unique values, and categories may not be meaningful
- Mix of categorical and numerical features can make statistical modeling difficult
- And many more

Learning objectives

At the end of this lecture, you should be able to

- Describe variables in a given dataset, including their number, types, distributions, and descriptive statistics.
- Examine relationships between variables using appropriate numerical summaries and visualizations.
- Identify and justify variables that may be useful for modelling the target variable.
- Use the EDA findings to **plan** the data processing needed to prepare the data for modelling.

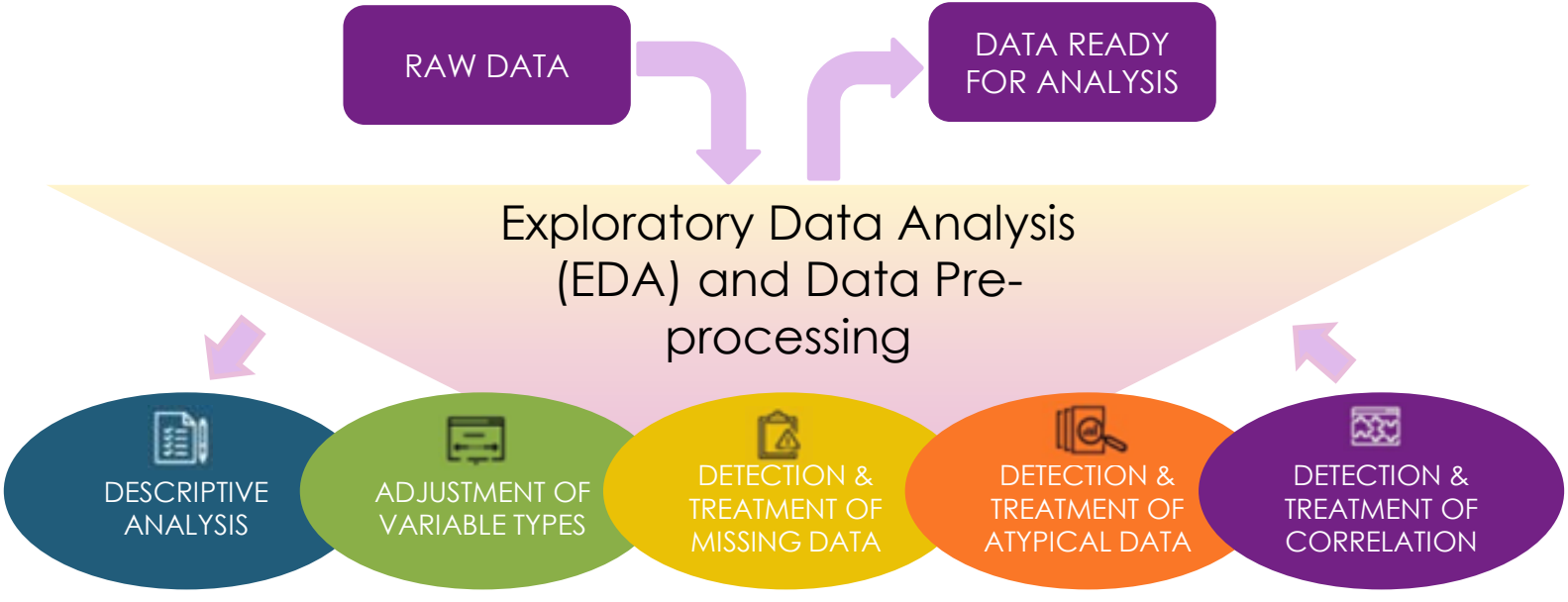
Exploratory data analysis (EDA)

- Exploratory data analysis (EDA) is the process to analyze and investigate data sets and summarize their main characteristics
    - How many variables do you have, what are their types, what is the distribution of data points? Are they comparable?
    - Are there issues such as missing values, extreme outliers, large deviations?
    - Are features related to each other?

Data pre-processing

- Data pre-processing can refer to manipulation, filtration or augmentation of data before it is analyzed
- Also known as data cleaning, data wrangling

EDA and pre-processing



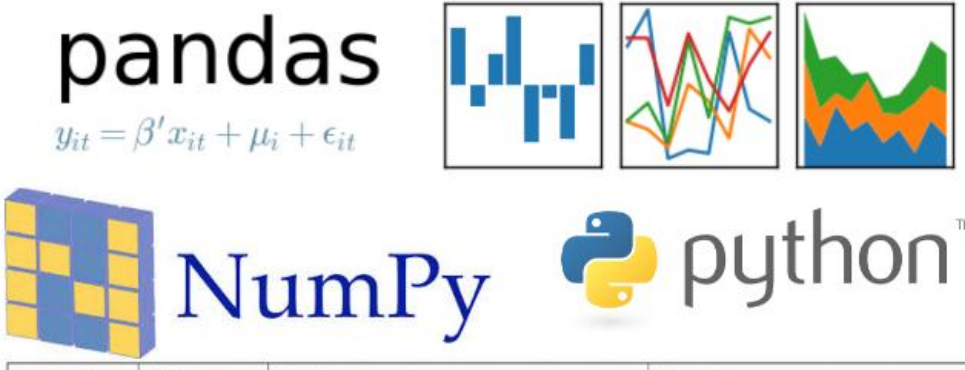
RECAP: Variable types

- Variable types
  - Nominal/Categorical: Category, code, state (e.g., name of a ship)
  - Binary: Nominal attribute with only 2 categories or states (e.g., fuel or electric)
  - Ordinal: Attribute with possible values that have a meaningful order or ranking among them (e.g., Class A, Class B, Class C type of vessels)
  - Numeric: A numeric attribute is quantitative; that is, it is a measurable quantity, represented in integer or real values (e.g., ship dimensions)

Data types

- A data type is essentially an internal construct that a programming language uses to understand how to store and manipulate data.

- When you use Pandas to read a (CSV) file into a DataFrame, Pandas automatically infers the data type of each column based on the values it contains. – This automatic assignment may be incorrect



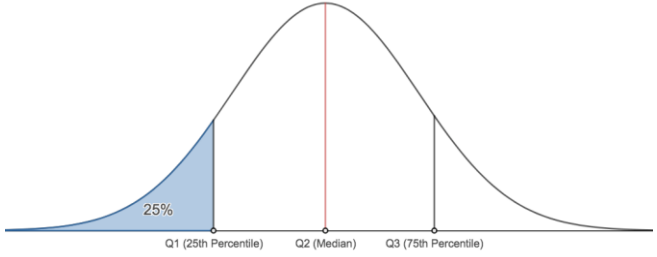
Pandas dtype	Python type	NumPy type	Usage
object	str or mixed	string_, unicode_, mixed types	Text or mixed numeric and non-numeric values
int64	int	int_, int8, int16, int32, int64, uint8, uint16, uint32, uint64	Integer numbers
float64	float	float_, float16, float32, float64	Floating point numbers
bool	bool	bool_	True/False values
datetime64	NA	datetime64[ns]	Date and time values
timedelta[ns]	NA	NA	Differences between two datetimes
category	NA	NA	Finite list of text values

Descriptive Statistics

- Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers.
- Descriptive statistics are statistics that summarize or describe variables of a data set
- Descriptive statistics consists of 3 basic categories of measures:
  - Measures of central tendency (mean, median, mode)
  - Measures of variability (variance, standard deviation, minimum and maximum, skewness)
  - Measures of frequency distribution (count)

Range, Quartile, Interquartile range

- The range is the difference between the largest (max()) and smallest (min()) values of a variable
- First quartile Q1 (25th percentile) – cuts off the lowest 25% of the data
- Third quartile Q3 (75th percentile) – cuts off the highest 25% of the data
- Second quartile Q2 (50th percentile) – As the median, it gives the center of the data distribution
- Interquartile range IQR = Q3 – Q1



Five number summary

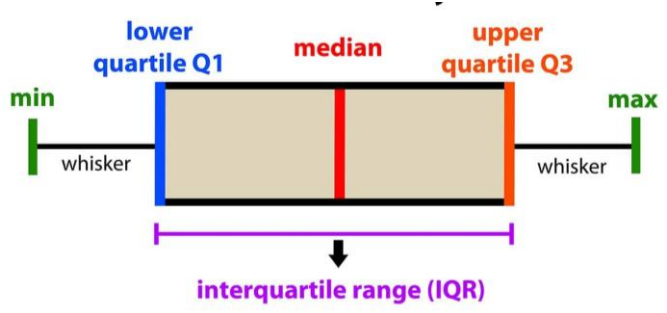
- Because Q1, the median, and Q3 together contain no information about the endpoints (e.g., tails) of the data, a fuller summary of the shape of a distribution can be obtained by providing the lowest and highest data values as well. This is known as the five-number summary.

- The five-number summary of a distribution consists of the median (Q2), the quartiles Q1 and Q3, and the smallest and largest individual observations, written in the order of Minimum, Q1, Median, Q3, Maximum.

Visualization – Boxplot

Boxplots are a popular way of visualizing the key characteristics of a distribution. A boxplot incorporates the five-number summary as follows:

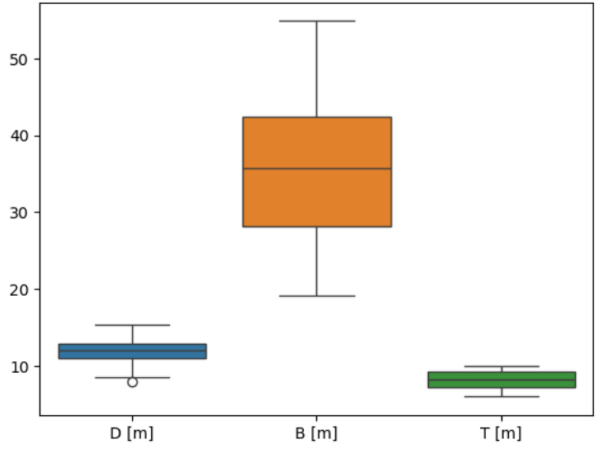
- Typically, the ends of the box are at the quartiles so that the box length is the interquartile range.
- The median is marked by a line within the box.
- Two lines (called whiskers) outside the box extend to the smallest (Minimum) and largest (Maximum) observations.



Visualization – Boxplot

Boxplots are a popular way of visualizing the key characteristics of a distribution.

- When dealing with a moderate number of observations, it is worthwhile to plot potential outliers individually.
- To do this in a boxplot, the whiskers are extended to the extreme low and high observations only if these values are less than 1.5 × IQR beyond the quartiles.



Reflection – Look at the following task in the assignment

Grade 3 to 4: Variation in the variables: Examine how the variables are distributed across the dataset. Explain the main patterns you observe, including their typical values, variability, and any notable skewness, imbalance, or unusual observations.

Correlation analysis

- Given two attributes, correlation analysis can measure how strongly one attribute implies the other, based on the available data.
- For categorical data, we can use the $\chi 2$ (chi-square) test.
- For numeric attributes, we can use the cross-moment, covariance or correlation coefficient, all of which access how one attribute's values vary from those of another.

Correlation coefficient

$$c_{XY} = E\left[(X - m_x)(Y - m_y)\right]$$

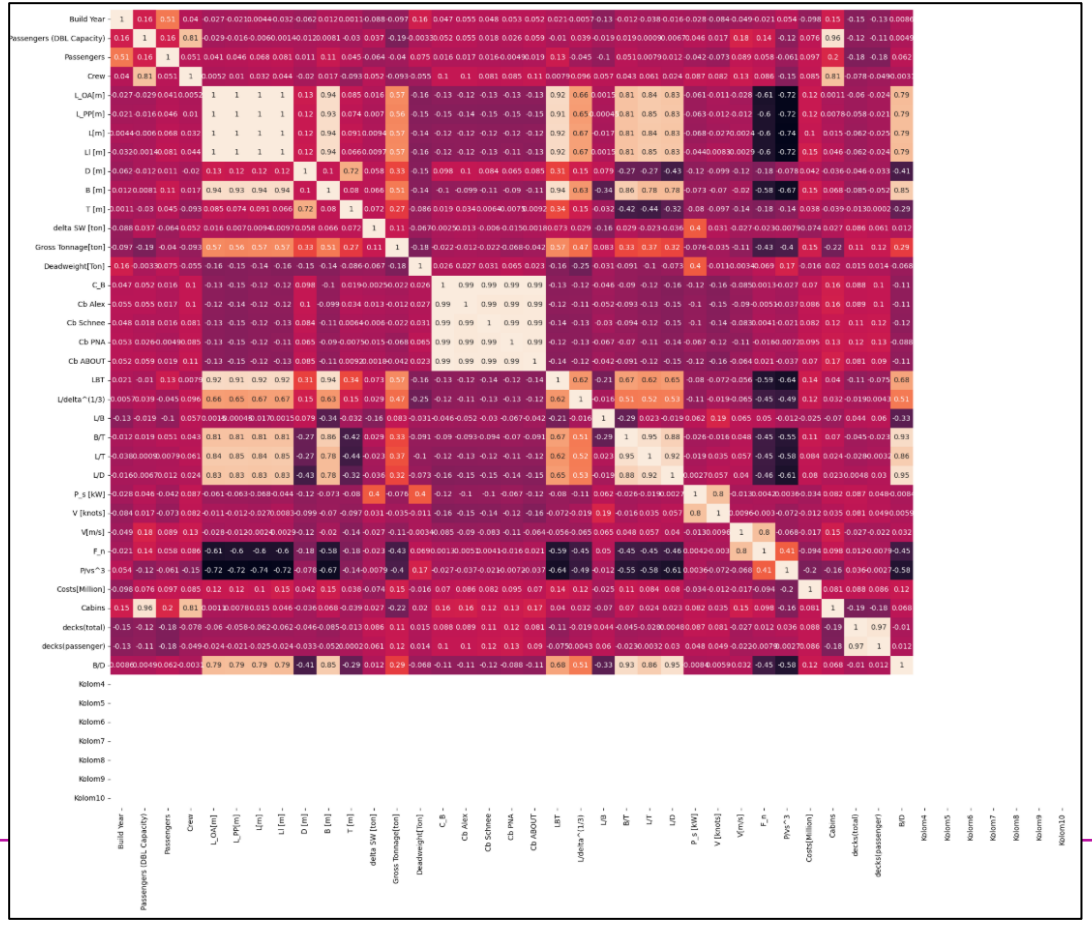
$$\boxed{\rho_{XY} = \frac{c_{XY}}{\sigma_X \sigma_Y}}$$

$$\sigma_X = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}}$$

- Property: $-1 \leq CorrCoeff \leq 1$
- $0 < CorrCoeff \leq 1$ means variables are positively correlated
- $-1 \leq CorrCoeff < 0$ means variables are negatively correlated

Correlation coefficient

- Explore the different correlation coefficient – Pearson, Spearman's
- When is the correlation coefficient a good measure of the relationship between variables? (Assumptions)



Visualization – Scatter plot

- Two attributes, X, and Y, are correlated if one attribute implies the other.
- Correlations can be positive, negative, or null (uncorrelated).
- The scatter plot is a useful method for providing a first look at bivariate data to see clusters of points and outliers, or to explore the possibility of correlation relationships.

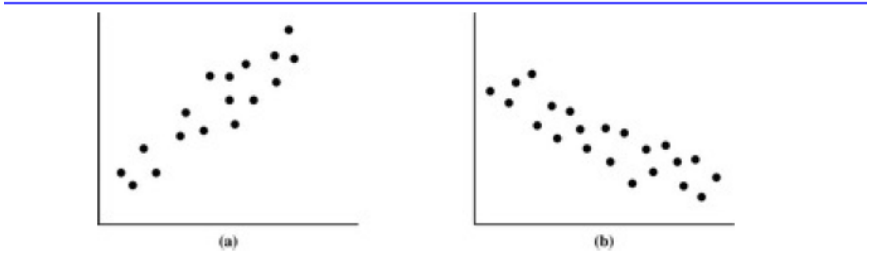
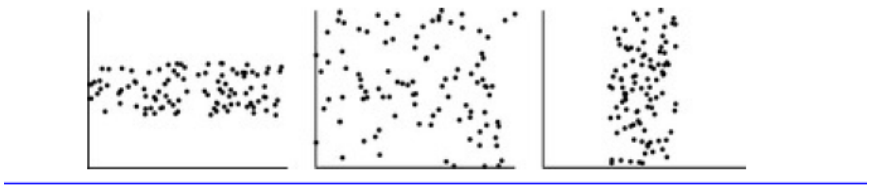


Figure 2.8 Scatter plots can be used to find (a) positive or (b) negative correlations between attributes.



e **2.9** Three cases where there is no observed correlation between the two plotted attributes in each of the data sets

Reflection – Look at the following task in the assignment

Grade 5: Variables relevant to propulsion shaft power: Investigate how propulsion shaft power relates to the other variables in the dataset. Also examine whether some variables are closely related to each other and may therefore provide similar information. Based on these relationships, identify which variables should be considered when modelling shaft power and explain your reasoning.

BREAK (5 minutes)

Exercise for today - 1

- Find a peer and work through the Python exercise together.
- Open this week's **EDA Numerical** notebook and make sure it is ready to run.
- The code is already provided. You only need to enter the correct CSV file path so that the notebook can read the **Cruiseships** data.
- Work through the notebook together: **uncomment and run the code, examine the results, and answer the discussion questions**. (~15 minutes)
- We will discuss selected questions together as a class afterwards. (~5 minutes)

BREAK (5 minutes)

RECAP: Python packages for data analysis – Visualization libraries



- Matplotlib (plots and graphs)
  - Most popular
  - Highly customizable



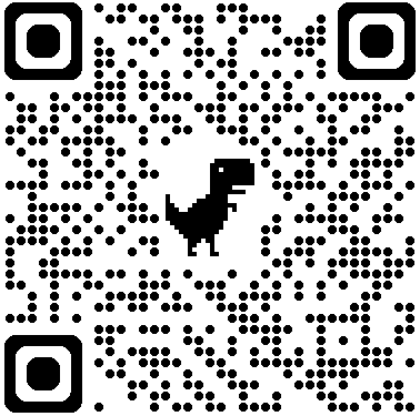
- Seaborn (Complex plots)
  - Based on Matplotlib
  - Heat maps, time series, violin plots

Exercise for today - 2

- Find a peer and work through the Python exercise together.
- Open this week's **EDA Visualization** notebook and make sure it is ready to run.
- The code is already provided. You only need to enter the correct CSV file path so that the notebook can read the **Cruiseships** data.
- Work through the notebook together: **uncomment and run the code, examine the results, and answer the discussion questions**. (~15 minutes)
- We will discuss selected questions together as a class afterwards. (~5 minutes)

Quiz

- Complete your quiz here.



Feedback

- Provide your feedback here

